Curriculum Vitae

Dr. Rachel Glenn
Medical Physics Fellow / Resident
Department of Imaging Physics
The University of Texas MD Anderson Cancer Center
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EDUCATION

M.S. **Medical Physics**
Department of Imaging, UTHealth / MD Anderson, 2024
Thesis: Hybrid Quantum/Classical Image Segmentation Algorithms
Committee: David T. Fuentes (chair), Gary V. Martinez, Jason Stafford, James A. Bankson

Ph.D. **Condensed Matter Physics**
Department of Physics, University of Utah, Salt Lake City, 2012
Thesis: Many-Body and Spin-Orbit Aspects of the Alternating Current Phenomena
Committee: M.E. Raikh (chair), Oleg Starykh, Christoph Boehme, Kyle Dawson, Elena Cherkaev

B.S. **Physics with Honors, Minor in Mathematics**
Weber State University, Ogden, 2008

B.S. **Electronics Engineering Technology**
Weber State University, Ogden, 2008

A.S. **Electronics Technology**
Salt Lake Community College, Salt Lake City, 2002

TEACHING

2026 **Imaging & Innovation Summer Immersion Program**
Nuclear Medicine Lab, UTHealth / MD Anderson

2020 **Electronics Laboratory**
Adjunct Professor, Department of Electrical and Computer Engineering, Baylor University

2015 **Special Topics in Physical Chemistry (CEM-988)**
Postdoctoral Scholar, Department of Chemistry, Michigan State University,

RESEARCH

2025- present **Medical Physics Fellow Resident**
Department of Imaging, UTHealth/MD Anderson

- Conduct translational research in quantitative MRI for precision oncology, focusing on diffusion MRI, multiparametric imaging, and biopsy-correlated radiomic biomarkers for glioma characterization.
- Develop and validate imaging biomarkers (ADC, IVIM, diffusion kurtosis, CTRW) and machine learning models to predict tumor cellularity and support image-guided diagnosis, adaptive radiotherapy, and treatment planning.
- Integrate quantitative imaging, pathology, and computational modeling to improve clinically interpretable biomarkers for MRI-guided cancer care.

2024-2025 **Associate Fellow / Senior AI & Machine Learning Engineer**
Lockheed Martin

- Developed advanced computational imaging algorithms for deployable synthetic aperture radar (SAR) systems, including image reconstruction, motion compensation, compressed sensing, and three-dimensional image formation.
- Designed AI-assisted reconstruction and image enhancement methods to improve image quality, robustness, and operational performance in challenging sensing environments.
- Collaborated with multidisciplinary engineering teams to translate novel computational imaging algorithms into deployable real-time imaging systems.
- Authored multiple internal research proposals in quantum computing for autonomous sensing systems, securing funding for next-generation computational imaging research.

2021-2024 **Research Associate / Graduate Student**
UTHealth / MD Anderson

- Developed deep learning and image segmentation methods for MRI-based radiation therapy planning and quantitative image analysis.
- Evaluated classical and quantum-inspired computational approaches for medical image segmentation, auto-contouring, and treatment planning.
- Contributed to the development and validation of MRI segmentation models for liver and abdominal imaging across diverse clinical datasets.
- Integrated artificial intelligence into clinical imaging workflows to improve reproducibility, efficiency, and quantitative analysis.

2017-2021 **Artificial Intelligence Architect**
Advanced Development Programs, Lockheed Martin

- Contributed to the technical leadership of multidisciplinary research in artificial intelligence, computational modeling, and advanced sensing technologies.
- Built international collaborations with universities, industry, and government to develop research programs in quantum technologies and advanced computing.
- Developed digital twin simulations and AI-enabled 3D environmental models for autonomous systems.

• Authored successful research proposals that secured funding for next-generation AI acceleration technologies.

2016-2017 **Senior Research Associate**
Department of Chemistry, Virginia Polytechnic Institute and State University
• Developed time-dependent quantum chemistry methods for modeling molecular response and spectroscopy.
• Designed computational algorithms for simulating light–matter interactions and electronic structure.

2014-2016 **Postdoctoral Researcher**
Department of Chemistry, Michigan State University
• Developed theoretical and computational methods for nonlinear optical spectroscopy.
• Collaborated closely with experimental imaging and spectroscopy groups to interpret complex multidimensional optical measurements.
• Created quantitative data analysis methods for ultrafast spectroscopy and optical imaging.

2013-2014 **Postdoctoral Scholar**
Department of Chemistry, University of California, Irvine
• Developed theoretical models of multidimensional optical spectroscopy using quantum and classical electromagnetic fields.
• Investigated nonlinear light–matter interactions, energy transfer, and pulse-shaping methods for advanced spectroscopic imaging.

2012 **Postdoctoral Researcher**
Department of Physics and Astronomy, University of Utah
• Investigated magnetic resonance phenomena through theoretical and experimental studies of spin dynamics and magnetic field modulation.
• Developed analytical models describing magnetic resonance response in condensed matter systems.

2010-2012 **Research Assistant**
Department of Physics and Astronomy, University of Utah
• Investigated spin-dependent transport, magnetic resonance, and light–matter interactions in semiconductor materials.
• Developed theoretical models describing electrically detected magnetic resonance (EDMR), spin-orbit interactions, and quantum transport.
• Applied advanced analytical methods to understand magnetic resonance phenomena relevant to condensed matter physics.

IMAGING & AUTOMATION SYSTEMS EXPERIENCE

2002-2012 **Airways Transportation System Specialist (ATSS), FV-H (2101)**
Interim appointment as Unix Administrator for STARS automation system (FV-I)

Federal Aviation Administration, Salt Lake City, UT

- Maintained, calibrated, upgraded, and certified mission-critical radar and automation systems operating in a 24/7 safety-critical environment.
- Performed system commissioning, acceptance testing, preventive maintenance, troubleshooting, and performance verification of complex electronic and imaging-related systems.
- Administered Unix-based automation infrastructure supporting real-time operational systems.
- Developed technical expertise in quality assurance, system reliability, fault analysis, and regulatory compliance.
- Collaborated with engineers, operators, vendors, contractors, and management to minimize downtime and maintain continuous system availability.
- Trained and certified technical staff through structured on-the-job training and competency evaluations.
- Earned journeyman certifications in Mode-S, ASR-9, ASDE-X, TDWR, LLWAS, and STARS radar and automation platforms.

INNOVATION & PROFESSIONAL DEVELOPMENT

- MIT Innovation Initiative Hackathon, MIT, Cambridge, MA (2019)
- LM Bright-Ideas (Sharktank) seedling funding AI deployed to FPGA (2019)
- LM Bright-Ideas (Sharktank) seedling funding atmostpheric simulations (2020)
- MIT Silicon Photonics Workshop, MIT, Cambridge, MA (2020)
- LM Bright-Ideas (Sharktank) seedling funding, Quantum Reinforcement Learning (2025)

PATENTS

2022 Method of Single Pulse Detection of the Real and Imaginary Components of the Nonlinear Susceptibility to Detect Molecular Dynamics via the Geometric Phase Effect, Publication No. 20200003681.

PUBLICATIONS

2026 Rachel Glenn, Tucker Netherton, Adrian Celaya, Mohamed Eltaher, Beatrice Riviere, Eugene J. Koay, David Fuentes. *Evaluation of quantum contouring algorithms for treatment planning on MR abdominal images*. Frontiers in Oncology. 2025;15:1553539. doi:10.3389/fonc.2025.1553539. Included in the 2026 Frontiers Research Topic, Emerging Fast Medical Imaging Techniques in Radiology.

2026 Rachel Glenn, Rajarajeswari Muthusivarajan, David A. Hormuth II, Max Wintermark, Ken Hwang, Dong-Eog Kim, David Fuentes, Dawid Schellingerhout. *Apparent Diffusion Coefficient (ADC) as a single predictor is insufficient to predict cellular density*. European Journal of Radiology 204 113119

2025 Rachel Glenn, Tucker Netherton, Adrian Celaya, Mohamed Eltaher, Beatrice Riviere, Eugene J. Koay, David Fuentes. *Evaluation of Quantum Contouring Algorithms for*

Treatment Planning on MR Abdominal Images. Frontiers in Oncology 15, 1553539.

2024 Awj Twam, Adrian E. Celaya, Megan C. Jacobsen, Rachel Glenn, Peng Wei, Jia Sun, Ann Klopp, Aradhana M. Venkatesan, David Fuentes, *Two-Stage Segmentation of Cervical Tumors Using PocketNet*. arXiv:2409.11456.

2024 Nihil Patel, Adrian Celaya, Mohamed Eltaher, Rachel Glenn, Kari Brewer Savannah, Kristy K. Brock, Jessica I. Sanchez, Tiffany L. Calderone, Darrel Cleere, Ahmed Elsaiey, Matthew Cagley, Nakul Gupta, David Victor, Laura Beretta, Eugene J. Koay, Tucker J. Netherton, and David T. Fuentes, *Training Robust T1-Weighted Magnetic Resonance Imaging Liver Segmentation Models Using Ensembles of Datasets with Different Contrast Protocols and Liver Disease Etiologies*, *Scientific Reports* 14, 20988.

2024 Adrian Celaya, Evan Lim, Rachel Glenn, Brayden Mi, Alex Balsells, Dawid Schellingerhout, Tucker Netherton, Caroline Chung, Beatrice Riviere, David Fuentes, *MIST: A Simple and Scalable End-to-End 3D Medical Imaging Segmentation Framework*, arXiv preprint arXiv:2407.21343.

2016 Ilyas Saytashev, Rachel Glenn, Galina A. Murashova, Saman Osseiran, David Spence, Cynthia L. Evans, and Marcos Dantus, *Multiphoton Excited Hemoglobin Fluorescence and Third Harmonic Generation for Non-Invasive Microscopy of Stored Blood*, *Biomed. Opt. Express* 7, 3449–3460.

2016 Rachel Glenn and Marcos Dantus, *Molecular Level Crossing and the Geometric Phase Effect from the Optical Hanle Perspective*, *Phys. Rev. A* 93, 043402.

2016 Rachel Glenn and Marcos Dantus, *Single Broadband Phase-Shaped Pulse Stimulated Raman Spectroscopy for Standoff Trace Explosive Detection*, *J. Phys. Chem. Lett.* 7, 117–125.

2015 Richa Mittal, Rachel Glenn, Ilyas Saytashev, Vadim V. Lozovoy, and Marcos Dantus, *Femtosecond Nanoplasmonic Dephasing of Individual Silver Nanoparticles and Small Clusters*, *J. Phys. Chem. Lett.* 6, 1638–1644.

2015 Rachel Glenn, Kochise Bennett, Konstantin E. Dorfman, and Shaul Mukamel, *Photon-Exchange Induces Optical Nonlinearities in Harmonic Systems*, *J. Phys. B: At. Mol. Opt. Phys.* 48, 065401.

2014 Rachel Glenn and Shaul Mukamel, *Nonlinear Transmission Spectroscopy with Dual Frequency Combs*, *Phys. Rev. A* 90, 023804.

2014 Rachel Glenn and Shaul Mukamel, *Multidimensional Spectroscopy with a Single Broadband Phase-Shaped Laser Pulse*, The Journal of Chemical Physics 140, 144105.

2014 Rachel Glenn, Mark E. Limes, Brian Saam, Christoph Boehme, and Mikhail E. Raikh, *Analytical Study of Spin-Dependent Transition Rates within Pairs of Dipolar and Strongly Exchange Coupled Spins with* S=1/2 *During Magnetic Resonant Excitation*, *Physical Review B* 87, 165205.

2013 Rachel Glenn, Mark E. Limes, Brendan Pankovich, Brian Saam, and Mikhail E. Raikh, *Magnetic Resonance in Slowly Modulated Longitudinal Field: Modified Shape of the Rabi Oscillations*, *Physical Review B* 87, 155128.

2013 Rachel Glenn, William Jacob Baker, Christoph Boehme, and Mikhail E. Raikh, *Analytical Description of the Spin-Rabi Oscillation Controlled Electron Transition Rates Between Weakly Coupled Pairs of Paramagnetic States with* S=1/2, *Physical Review B* 87, 155208.

2012 Rachel Glenn, Oleg A. Starykh, and Mikhail E. Raikh, *Interplay of Spin-Orbit Coupling and Zeeman Splitting in the Absorption Lineshape of 2D Fermions*, *Physical Review B* 86, 024423.

2011 Rachel Glenn and Mikhail E. Raikh, *Rabi-Vibronic Resonance with Large Number of Vibrational Quanta*, *Physical Review B* 84, 195454.

PROFESSIONAL AFFILIATIONS

2015-present	American Physical Society (APS)
2021-present	American Association of Physicists in Medicine (AAPM)

PROFESSIONAL PRESENTATIONS (Poster presentations indicated)

Aug. 19, 2026	Quantum Computing in The Next Five Years Joint NucMed Seminar
Aug. 25, 2026	Mechanistic Evaluation of Diffusion MRI for Predicting Glioma Cell Density (Poster) CCSG Image-Guided Interventions & Insights (III) Program Retreat
Jul. 19, 2026	Mechanistic Evaluation of Diffusion MRI for Predicting Glioma Cell Density (Poster) American Association of Physicists in Medicine
Mar. 28, 2026	Invited Workshop Presentation and Career Panel, Reach for the stars STEM Festival Rice University
Mar. 24, 2026	Quantum Algorithms for Medical Imaging: Motivation, Measurements, and Limits Rice University QS^3 Invited speaker

Aug. 8, 2025	AI-Enhanced Compressed Sensing for Synthetic Aperture Radar (SAR) Imaging Lockheed Martin Conference
May 15, 2024	Review of Quantum/Classical Image Processing Algorithms for Medical Images Society for Thermal Medicine
Apr. 2, 2024	Review of Quantum/Classical Image Processing Algorithms for Medical Images Thesis Defense
Mar. 21, 2024	Development of a Segmentation Model for Liver on Magnetic Resonance Imaging Tumor Measurement Initiative Symposium
Jul. 26, 2023	A Weighted Normalized Boundary Loss for Reducing the Hausdorff Distance in Medical Imaging Segmentation (Poster) American Association of Physicists in Medicine
Jul. 26, 2023	Quantum and Classical Image Processing Algorithms for Medical Imaging (Poster) American Association of Physicists in Medicine
Nov. 5, 2022	Incorporating GTV Information in a Multi-stage Process to Improve Automatically Generated Field Apertures for Rectal Cancer Radiotherapy (Poster) Society of Industrial and Applied Mathematics
Oct. 6, 2020	Silicon Photonics for Quantum Computing Hardware Lockheed Martin Lunch and Learn
May 5, 2020	Artificial Intelligence Agent Architectures Lockheed Martin Lunch and Learn
Aug. 13, 2019	Swarming for Collaborative Agent Systems (Poster) Lockheed Martin Conference
Mar. 5, 2019	Mapping of the Excited State Potential Energy Surface during Molecular Photoisomerization to Control Chemical Reactions APS March Meeting
Nov. 5, 2018	Conceptual Insights into the Interaction of the Electric Field and Molecule in Spectroscopy to Further Our Understanding of Molecular Dynamics such as Isomerization Lockheed Martin Lunch and Learn

Mar. 5, 2018	Conceptual Insights into the Interaction of the Electric Field and Molecule in Spectroscopy to Further Our Understanding of Molecular Dynamics Such as Isomerization APS March Meeting
Nov. 30, 2017	Introduction to Modern Quantum Chemistry and Real-Time Molecular Simulations Lockheed Martin Lunch and Learn
Oct. 19, 2017	Review of Real-Time Molecular Simulations Results and Project Turnover Crawford Research Group
Aug. 31, 2017	Localization Errors and Controllable Sparsity in the Coupled-Cluster Wave Function for Time- Dependent Linear Response Theory (Poster) WATOC 2017
Mar. 13, 2017	Time-Dependent Molecular Response with Both Electric and Magnetic Fields APS March Meeting
Aug. 19, 2016	Molecular Level-Crossing and Organic Semiconductors Microsystems Technology Office, Booz Allen Hamilton
Aug. 3, 2016	Measuring the Real and Imaginary Parts of the Molecular Polarization using a Single-Shaped Pulse Crawford Research Group, Virginia Polytechnic Institute and State University
Jul. 14, 2016	Molecular Level-Crossing and the Geometric Phase Effect from the Optical Hanle Perspective Wilson Research Group, Michigan State University
Mar. 16, 2016	Molecular Level-Crossing and the Geometric Phase Effect from the Optical Hanle Perspective APS March Meeting
Jun. 30, 2015	Multidimensional Stimulated Emission Spectroscopy with a Single Phase-Shaped Pulse TRVS 2015
Mar. 5, 2015	Femtosecond Nanoplasmonic Dephasing of Individual Silver Nanoparticles APS March Meeting
Mar. 4, 2015	Coherent Dynamics Following Strong Field Ionization of Polyatomic Molecules APS March Meeting

Mar. 3, 2015	Solvation Dynamics with a Single Phase-Shaped Laser Pulse (Poster) APS March Meeting
Apr. 4, 2014	Rabi-Vibronic Resonance and Photon-Exchange Induced Nonlinearities in Harmonic Oscillators Physics Colloquium, University of California, Merced
Mar. 8, 2013	Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors Naval Research Laboratory, Washington, DC
Nov. 17, 2012	Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors MRSEC (Engineering/Physics Department) University of Utah
Oct. 4, 2012	Spin (S=1⁄2) Rabi Spectroscopy of Weakly and Strongly Coupled Spin Pairs in Organic Semiconductors Quantum Information Processing, Physics Department University of Wisconsin-Madison
Mar. 20, 2012	Rabi-Vibronic Resonance with a Large Number of Vibrational Quanta Physics Department, University of Utah
Jan. 24, 2012	Peculiarities of the Rabi Oscillations in a Quantum Dot Molecule Physics Department, University of Utah